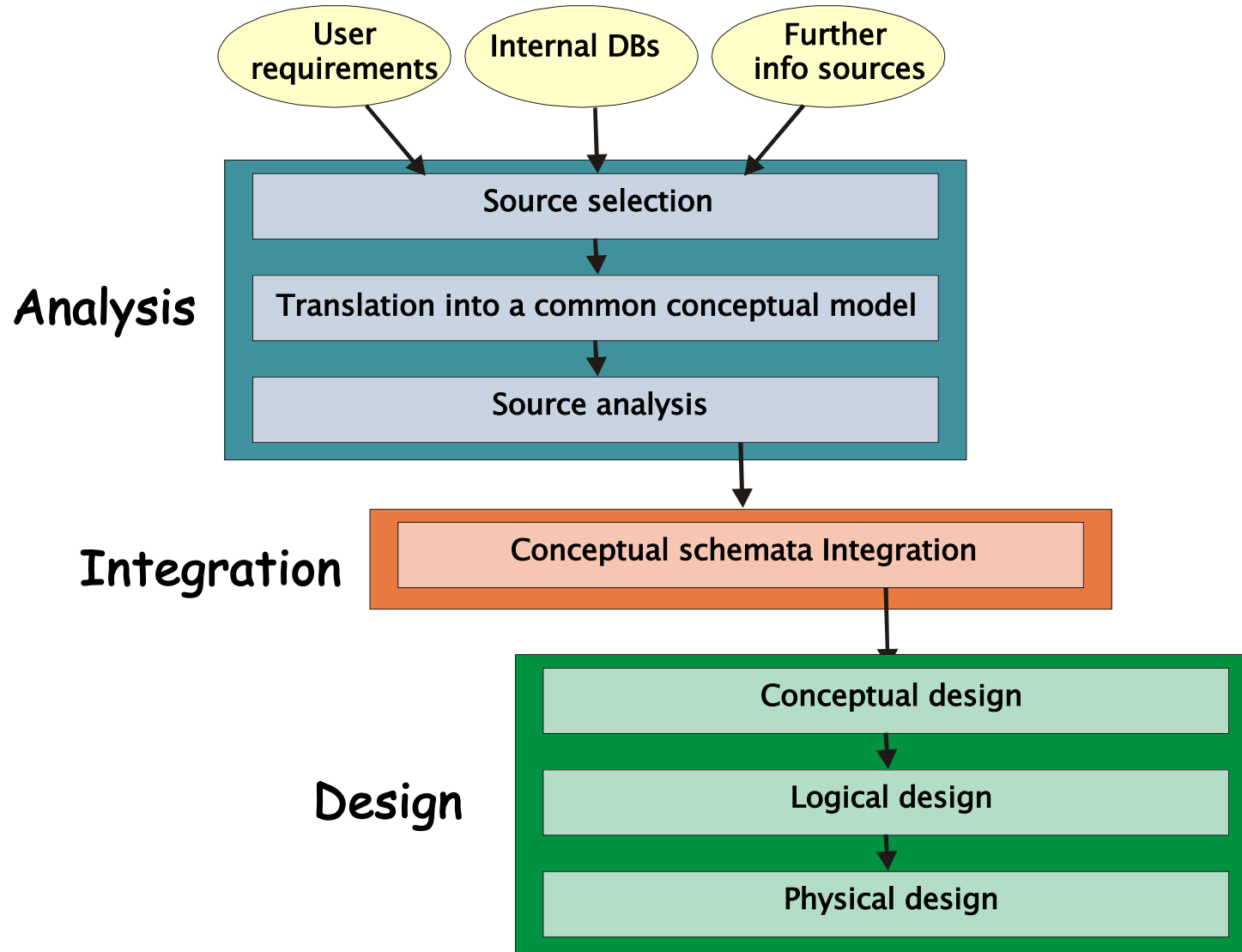


Data Warehouse Design

Letizia Tanca
Politecnico di Milano

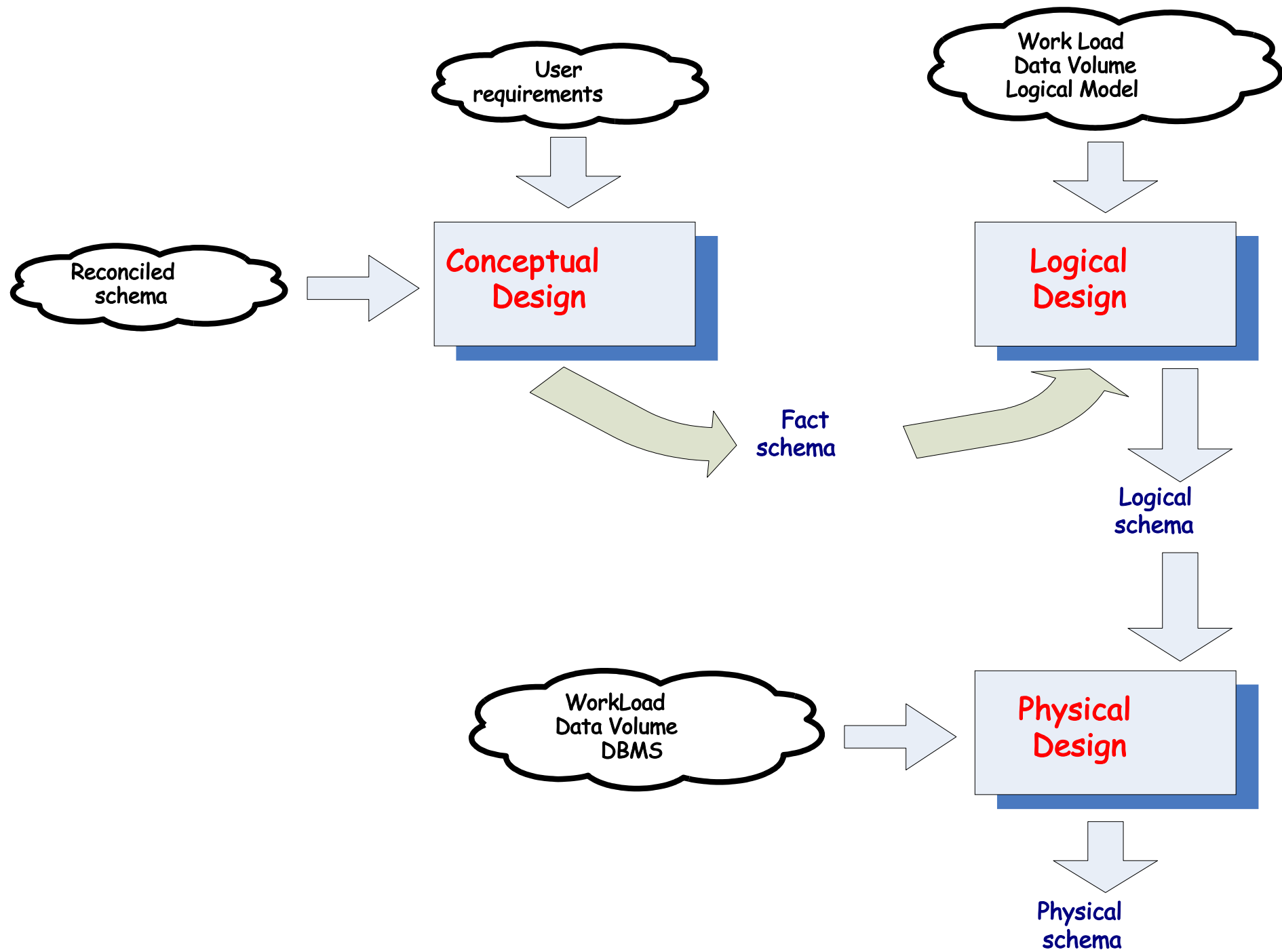
with kind thanks to
Rosalba Rossato

Data Warehouse Design



Data Warehouse Design

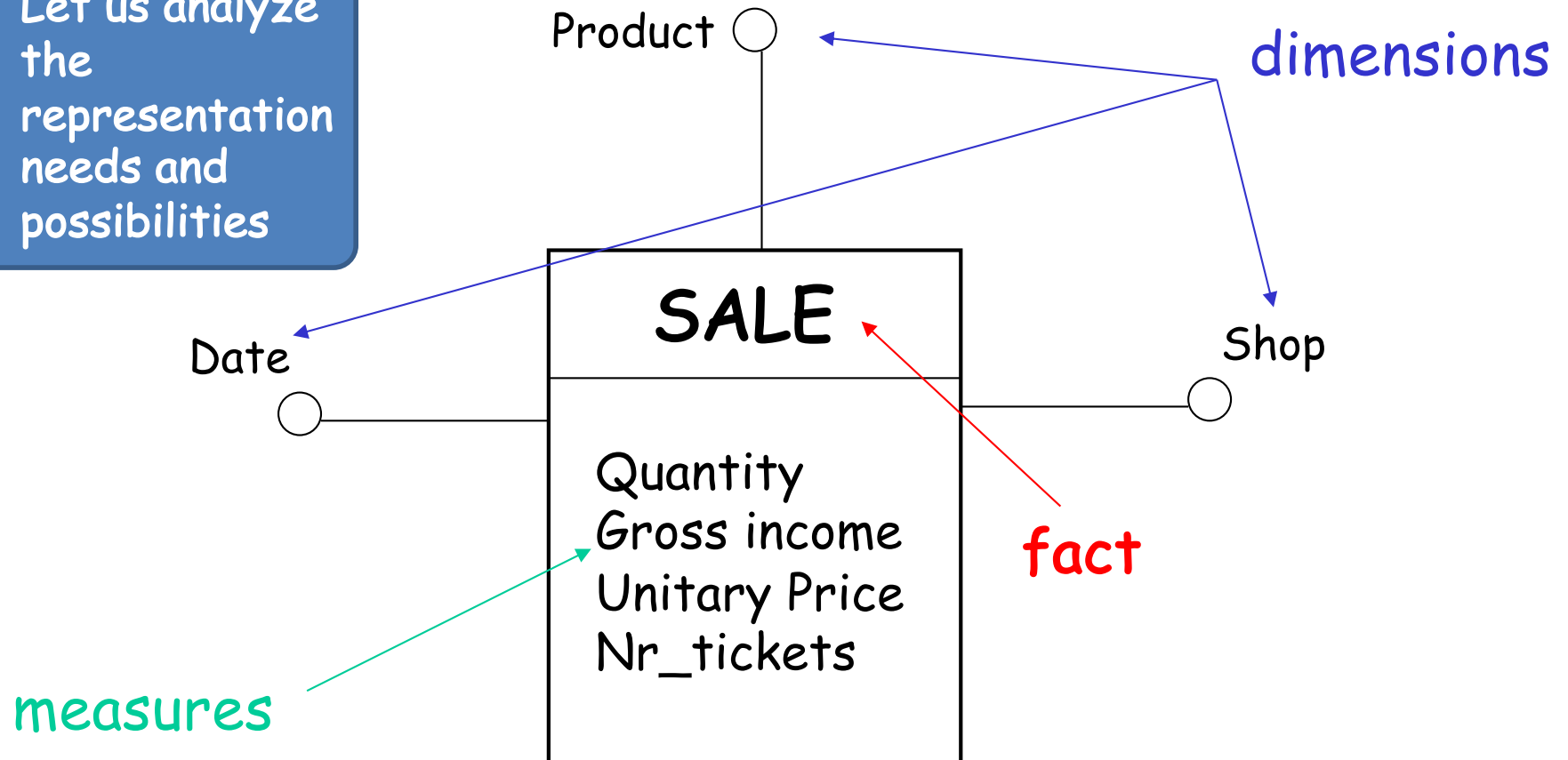
- Data Warehouses are based on the *multidimensional model*
- A common conceptual model for DW does not exist
- The Entity/Relationship model cannot be used in the DW conceptual design



Conceptual Modeling

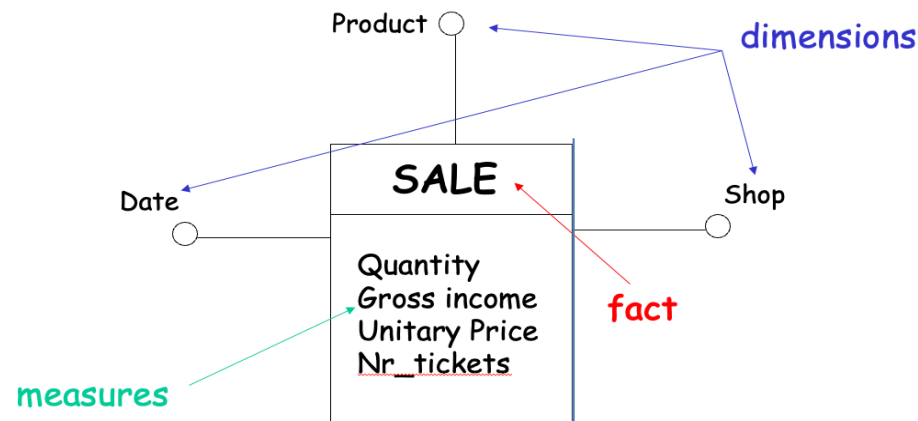
The Dimensional Fact Model (DFM)

Let us analyze the representation needs and possibilities

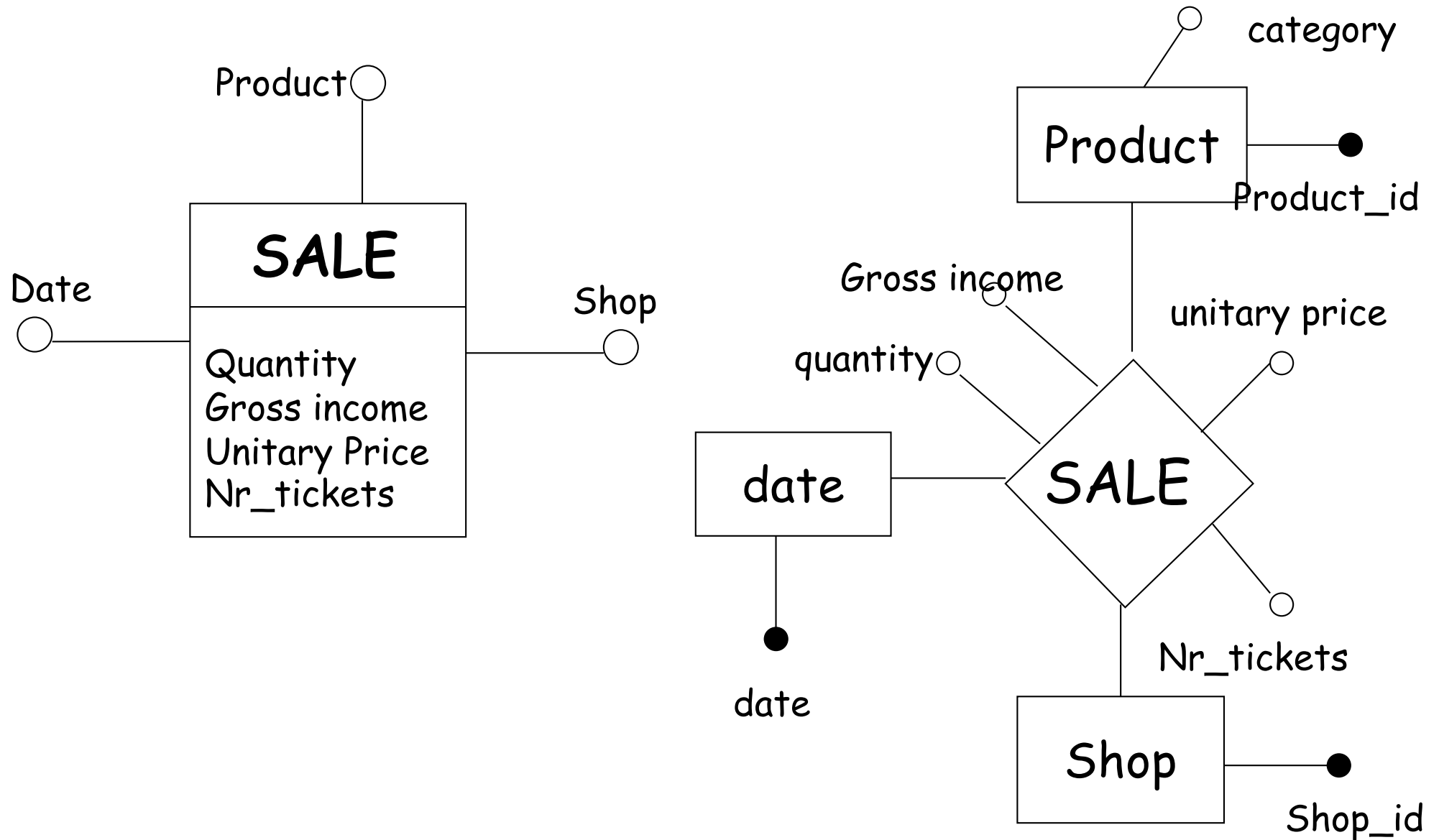


DFM Constraints

- A **fact** describes an **N:M relationship** among its **dimensions**
- There must be a **functional dependency** between the fact and its dimensions
- **Naming convention**: the dimensions of a same fact schema **must have distinct names**



DFM and E/R



Dimensional attribute

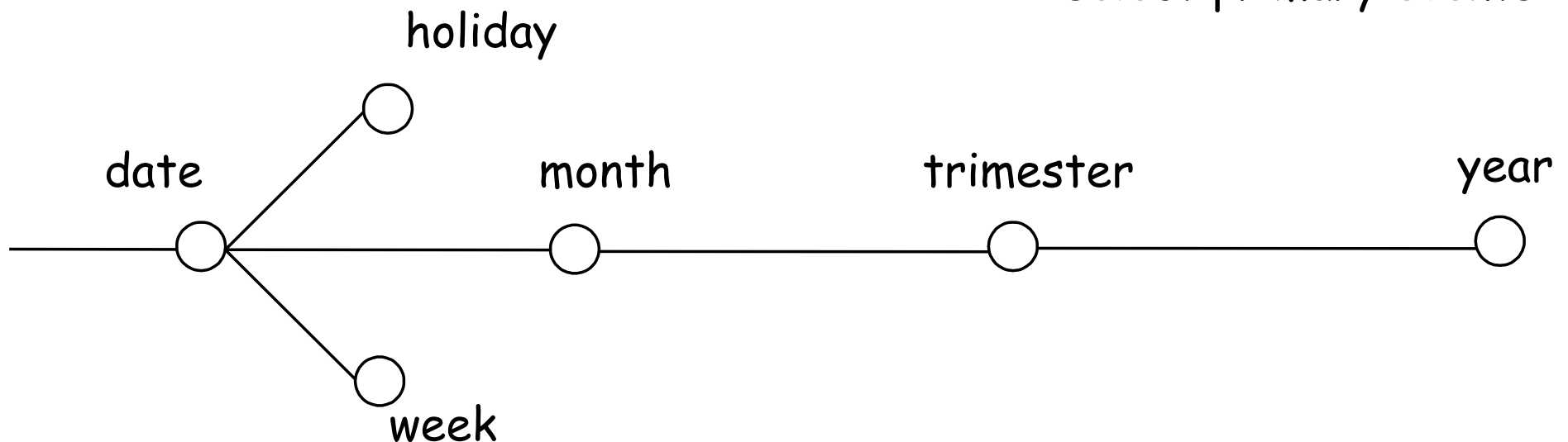
- A **dimensional attribute** must assume discrete values, so that it can contribute to represent a dimension
- Dimensional attributes can be organized into **hierarchies**

Hierarchies

A **dimensional hierarchy** is a directed tree whose

- **Nodes** are dimensional attributes
- **Edges** describe **n:1 associations** between pairs of dimensional attributes
- **Root** is the considered dimension

- The root of a hierarchy corresponds to the **primary events**, and represents the **finest aggregation granularity**
- The hierarchy describes how it is possible to group and select primary events

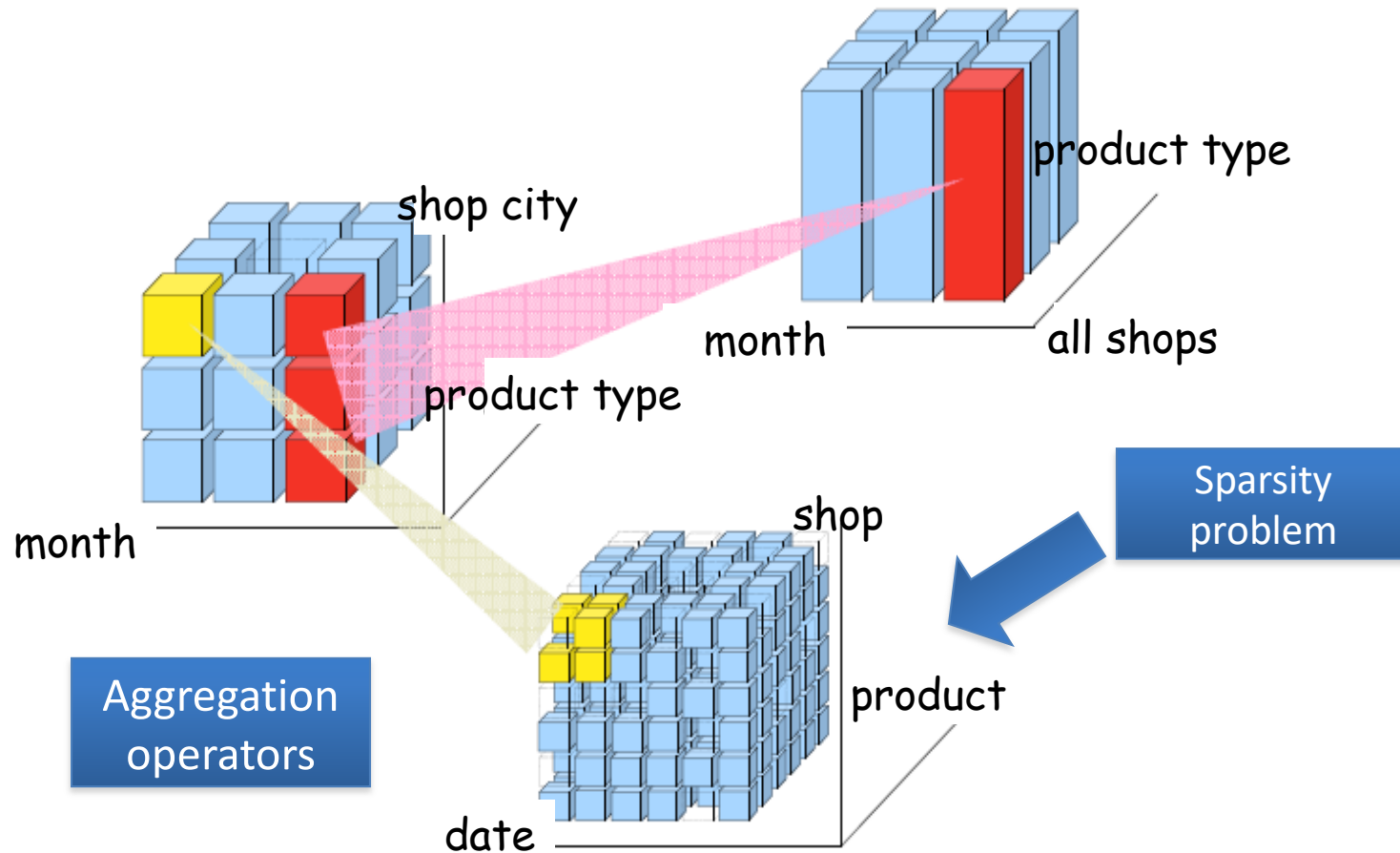


Events and aggregations

- A **primary event** is an occurrence of a fact; it is represented by means of a tuple of values:
On 10/10/2001, ten 'Brillo' detergent packets were sold at the BigShop for a total amount of 25 euros
- A **secondary event** aggregates some of the corresponding primary events: for example the sales can be grouped by Product and Month:

In October 2001, 230 'Brillo' detergent packets were sold at the BigShop for a total amount of 575 euros

Events and aggregations



Secondary event

- The sales can be further grouped by Product, Month, and City
- If we consider the dimensional attributes city, product and month, the tuple

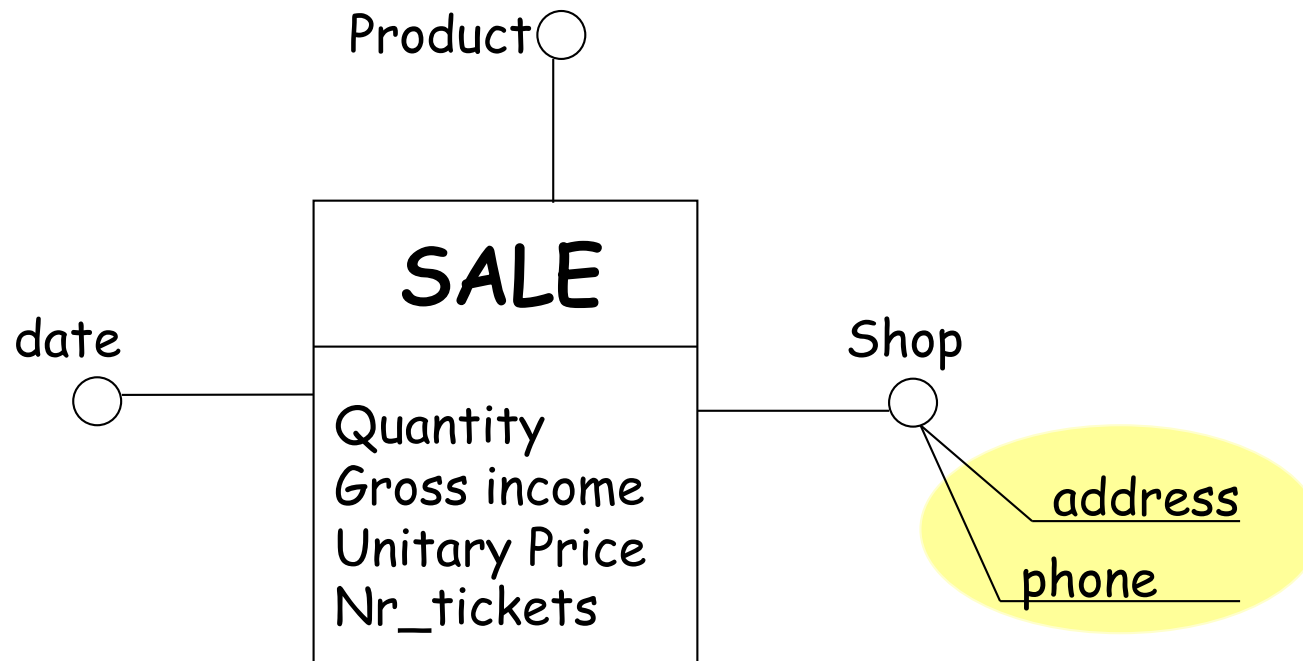
(city: 'Rome', product: 'Brillo', month: 10/2001)

identifies a secondary event

- It aggregates all the sales related to the product 'Brillo' in shops of 'Rome' during the month October 2001

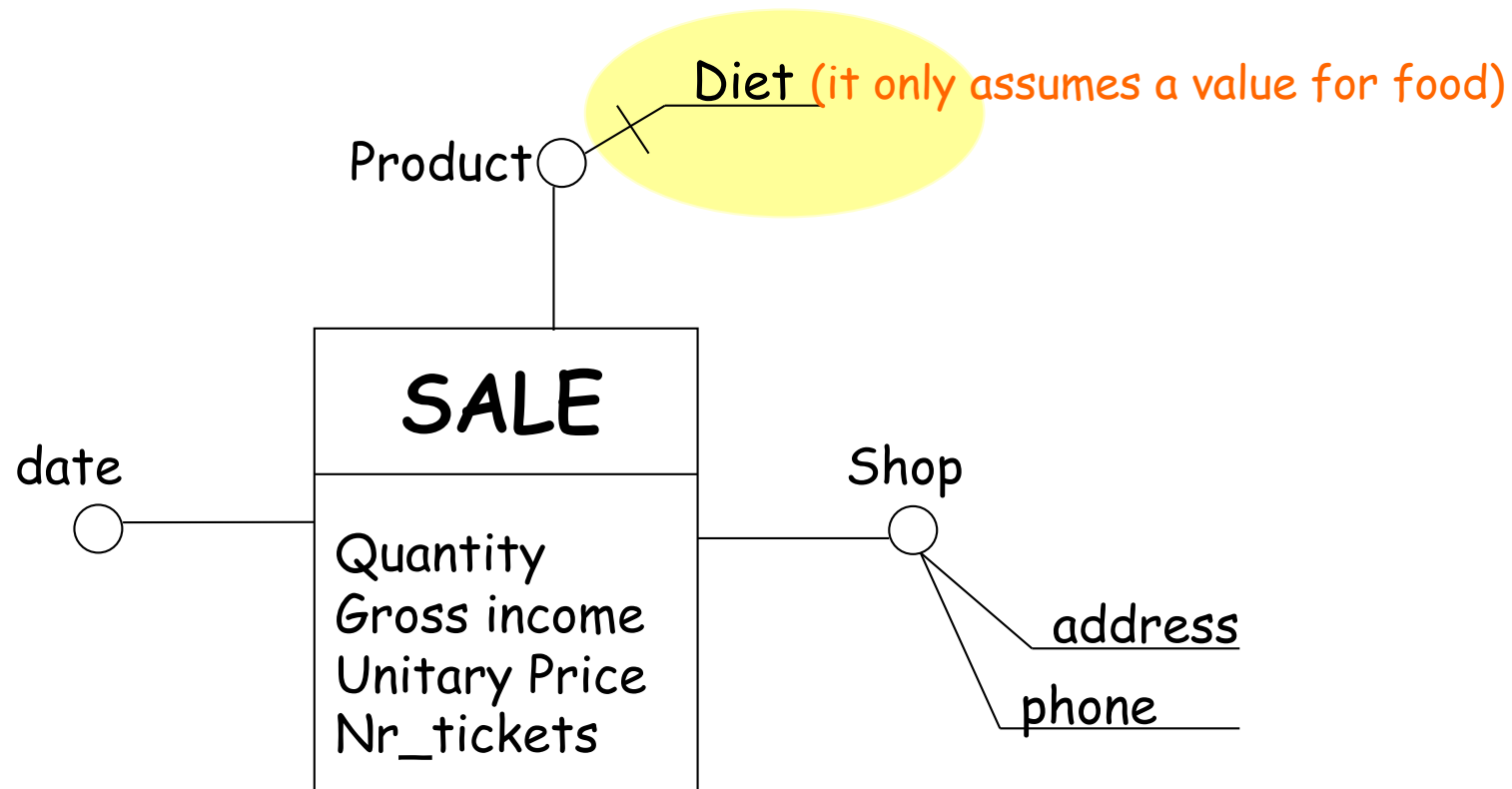
Descriptive attributes

- A **descriptive attribute** contains additional information about a dimensional attribute
- They are uniquely determined by the corresponding dimensional attribute



Optional edges

- Some edges of a fact schema could be optional



Optional dimensions



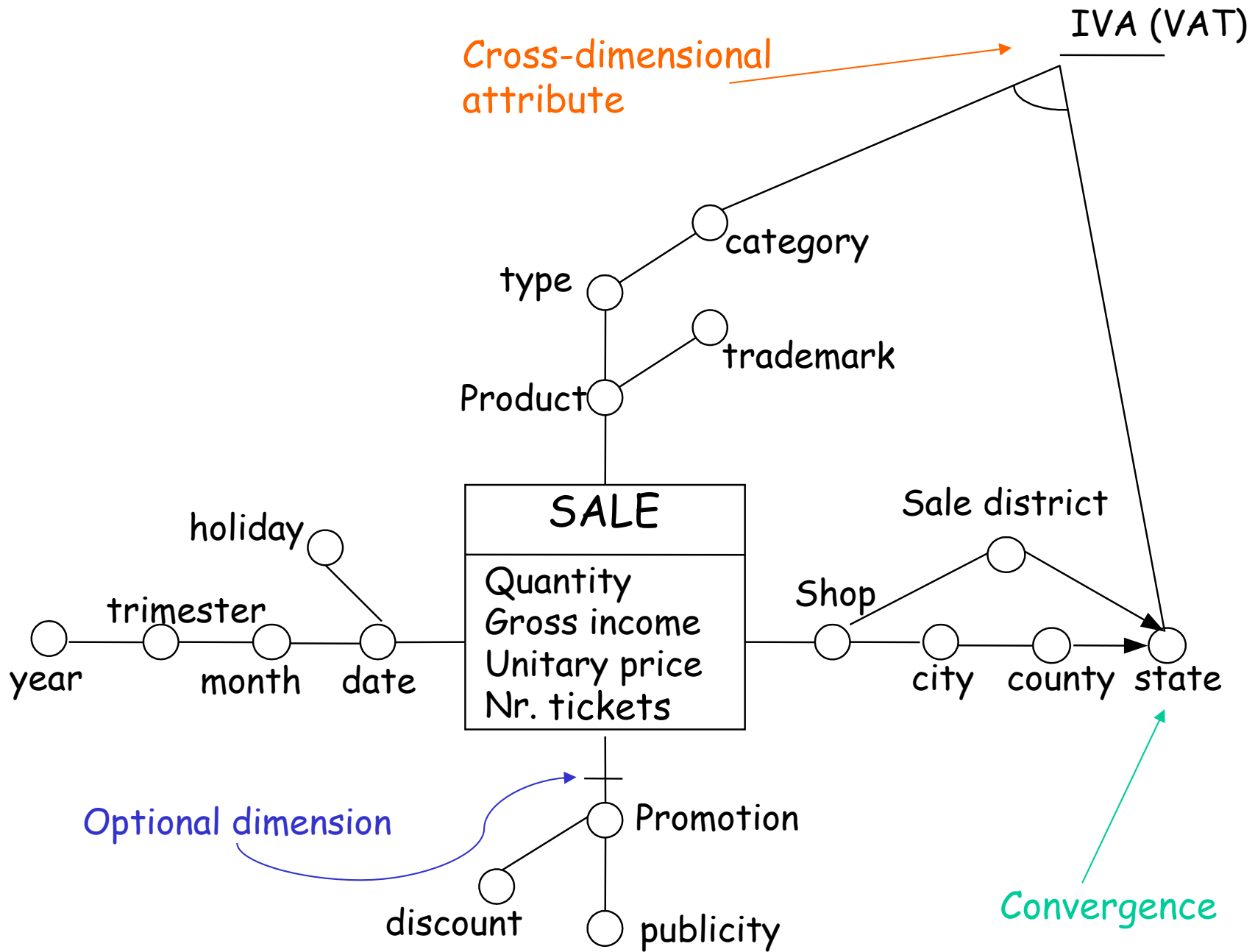
The attribute Promotion amount only assumes a value for products in promotion, therefore only products in promotion can be analysed along that dimension

Cross-dimensional attributes

- A **cross-dimensional attribute** is a dimensional or a descriptive attribute whose value is obtained by combining values of some dimensional attributes
 - ✓ For example, **IVA** (VAT) is computed based on the **product category** and the **state**

Convergence

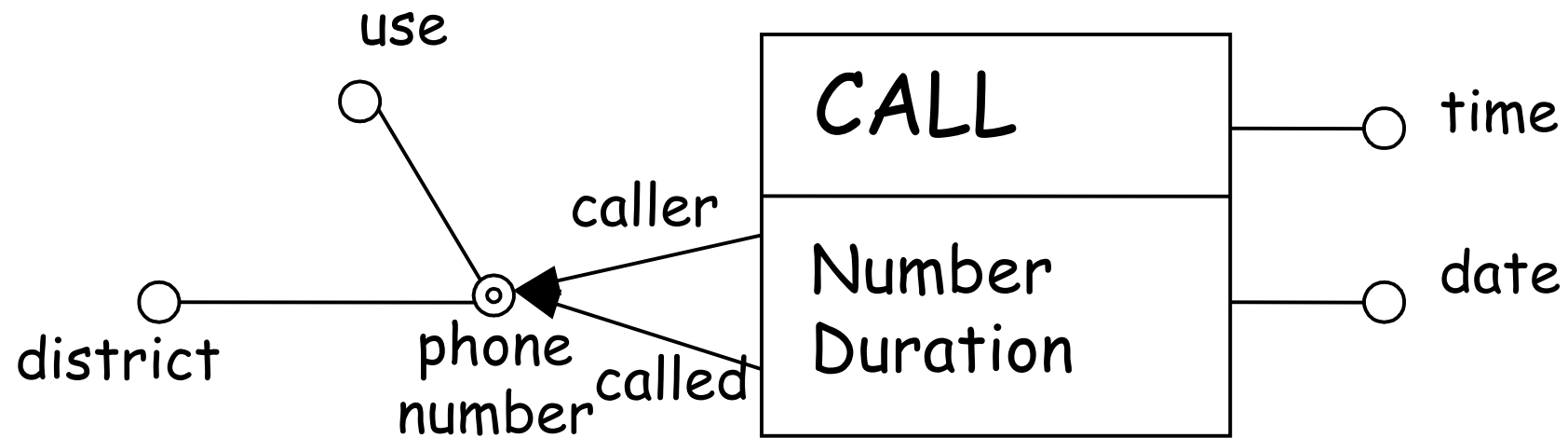
- It is related to the structure of a hierarchy
 - ✓ Two dimensional attributes may be connected by more than two distinct directed edges
 - ✓ For example:
 - Shop → city → county → state
 - or
 - Shop → sale district → state



Hierarchy Sharing

- In a fact schema, some portions of a hierarchy might be duplicated
- As a shorthand we allow **hierarchy sharing**
- If the sharing starts with a dimension attribute, it is necessary to indicate the **roles** on the incoming edges
- Necessary condition: the **functional dependency must hold on both branches**

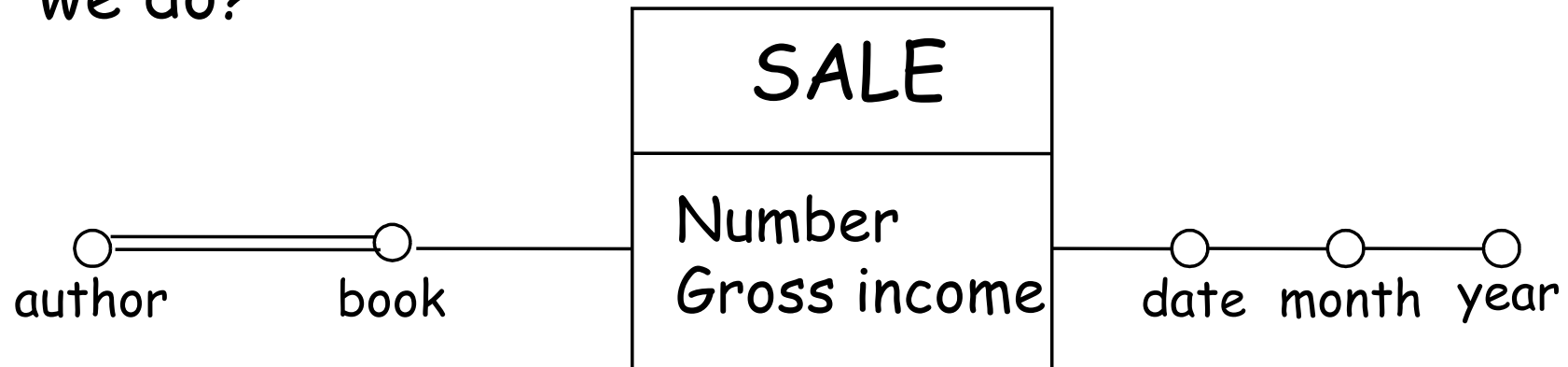
Hierarchy Sharing



It is in fact a shorthand to represent the duplication of the whole hierarchy

Multiple edges

- Recall: there must be a functional dependency between the fact and each dimension
- But some attributes, or some dimensions, may be related by a many-to-many relationship!!! What do we do?

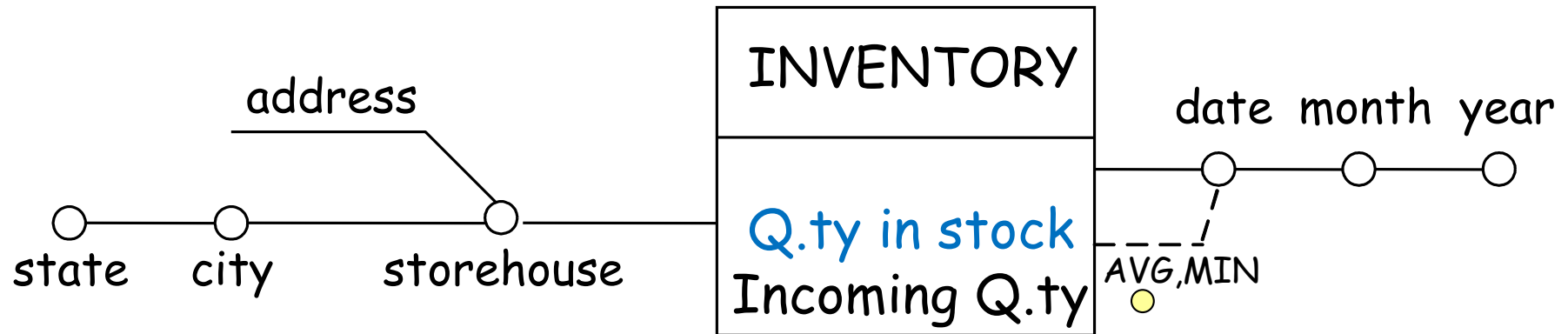


- At the conceptual level, we denote them by multiple edges, and at logical design time they are dealt with in a special way

Measure Aggregation

- Aggregation requires to specify an operator to combine the values related to primary events into a unique value related to a secondary event (e.g. sum of sold quantity aggregated by month)
- It is a rather critical subject
- A measure is **additive w.r.t. a given dimension** iff the **SUM** operator is freely applicable to that measure along that dimension
e.g., **amount sold** is additive w.r.t. **time and space**

Aggregability



This arc means that the **measure Q.ty in stock** (a so-called level measure) is **non-additive w.r.t. the time dimension**, but it is aggregable by means of the AVG and MIN operators

Measure Aggregation

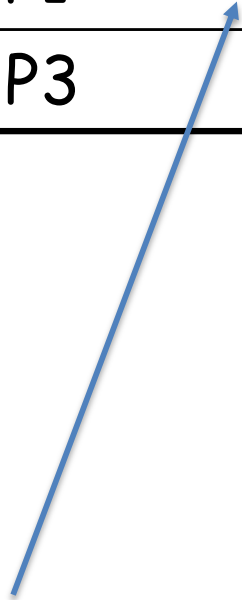
- Aggregation requires to specify an operator to combine the values related to primary events into a unique value related to a secondary event (e.g. sum of sold quantity aggregated by month)
- A measure is **additive w.r.t. a given dimension** iff the SUM operator is freely applicable to that measure along that dimension

e.g., amount sold is additive w.r.t. time and space while Q.ty in stock is not additive w.r.t. time

The n. of tickets is non-additive (and in general non-aggregable) w.r.t. the product

- By n. of tickets we mean how many times a product has been bought, i.e. **in how many tickets it appears**
- The association between product and ticket is **many-to-many**: a given product may be **in many tickets** and a ticket may contain **more than one product**
- E.g. by summing up the ticket count over the product type **we count the same type twice** if the ticket contains more than one product of same type

Ticket	Product	Type
S1	P1	T1
S1	P2	T1
S2	P1	T1
S2	P3	T2



Tickets containing p1 $\rightarrow 2$

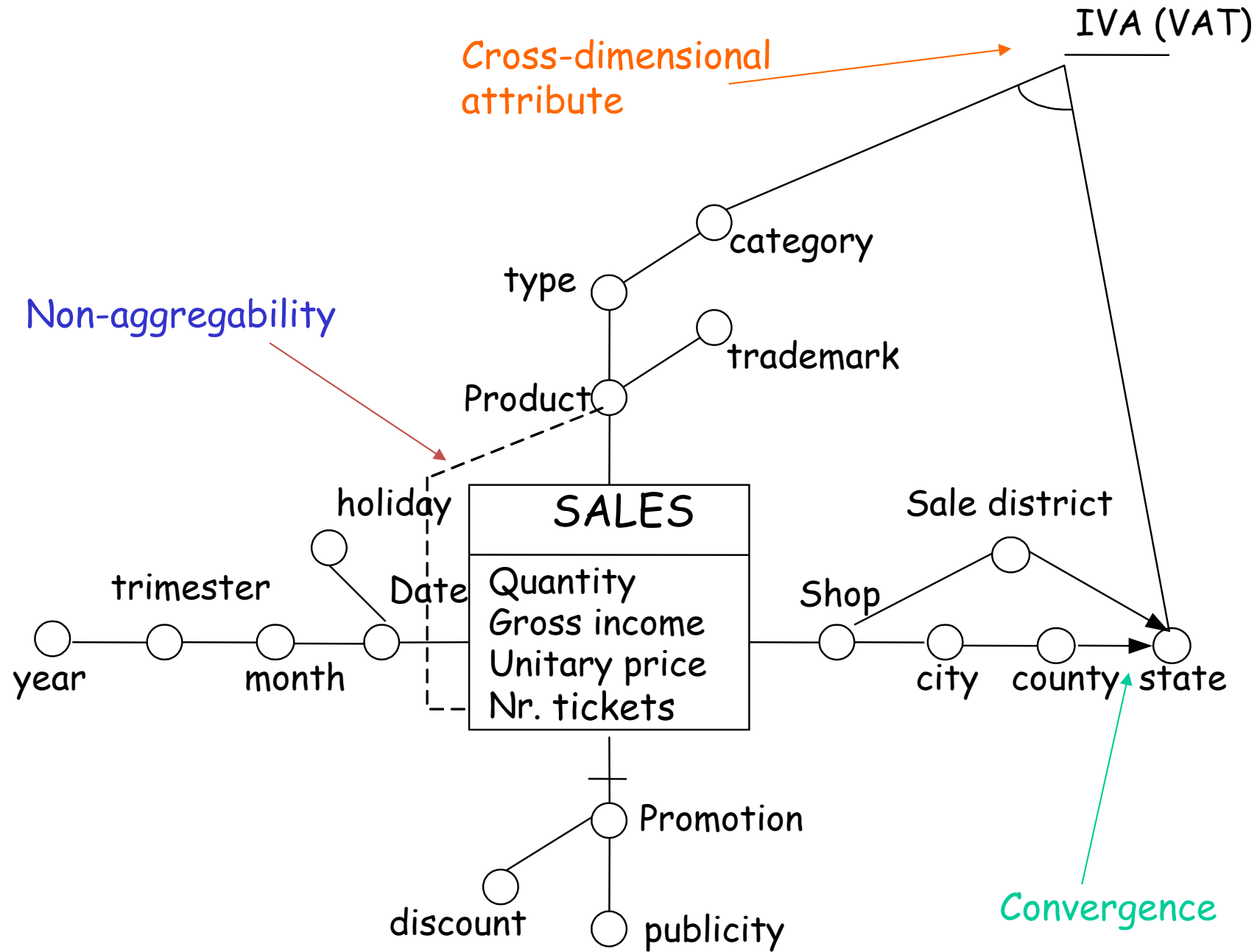
Tickets containing p2 $\rightarrow 1$

Tickets containing p3 $\rightarrow 1$

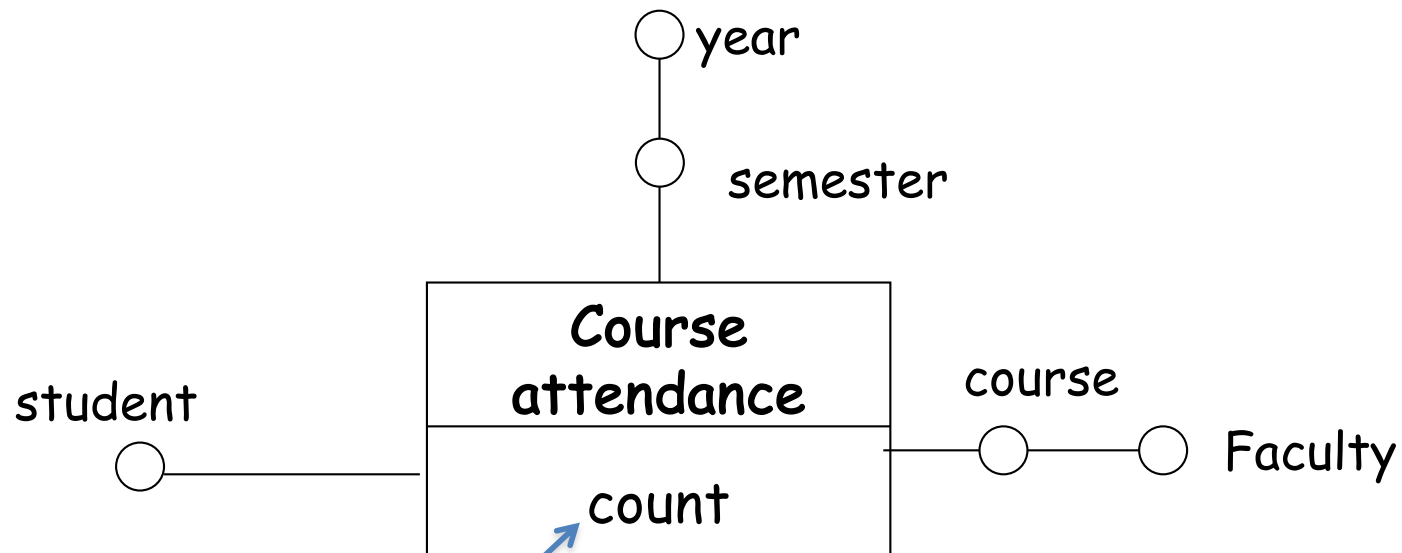
Tickets containing products of type t1 $\rightarrow 2$

HOWEVER, IF WE APPLY THE SUM
DIRECTLY TO THE TYPES:

Sum (tickets with Type(product) = T1) = 3 !!!



Empty fact schemata



We say that a fact schema is **empty** if there are no measures. In fact, the default measure is the **count**